

UNIT CONVERSION APPLICATION REPORTS



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CSA1021 INTRODUCTION TO PROBLEM SOLVING AND PROGRAMMING

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SEMESTER: Fall Semester 2026-27

PAGE 2 – CERTIFICATE

CERTIFICATE

This is to certify that **Akanksha Upadhyay**, Registration No.

26BAC10041, student of **ECE (AI&Cybernetics)**, has successfully completed the project entitled “**Unit Conversion Application**” as a part of the academic requirements for the session **2026– 27**.

The project has been completed under the guidance of the concerned faculty member. The work presented in this report is based on the implementation of a Python-based unit conversion application.

PAGE 3 – DECLARATION

DECLARATION

I hereby declare that the project entitled “**Unit Conversion Application**” has been prepared by me as part of my academic work.

The project is developed using the Python programming language. I have made sincere efforts to understand the concepts used in the project and implement them correctly.

I also declare that the information presented in this report is prepared for academic purposes.

Name: Akanksha Upadhyay

PAGE 4 – ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teachers and faculty members for their valuable guidance and support during the development of my project, **“Unit Conversion Application.”**

I am thankful to my college for providing me with the opportunity to work on this project. This project helped me improve my understanding of Python programming and its practical applications.

I would also like to thank my friends and family for their encouragement and support throughout the completion of this project.

Finally, I am grateful to everyone who directly or indirectly helped me in completing this project successfully.

PAGE 5 – ABSTRACT

ABSTRACT

The **Unit Conversion Application** is a Python-based application developed to perform different types of unit conversions quickly and easily.

The application provides a menu-driven interface where the user can

select a particular conversion category, enter the required value, and obtain the converted result.

The project includes several categories such as **Length, Weight, Temperature, Time, Area, Volume, Speed, and Digital Storage.**

The main purpose of this project is to demonstrate the practical implementation of Python programming concepts such as variables, input/output, conditional statements, loops, functions, and mathematical calculations.

The application is simple, useful, and suitable for students and beginners who want to understand programming through a real-world application.

PAGE 6 – INTRODUCTION

INTRODUCTION

Unit conversion is an important part of mathematics, science, engineering, and everyday life. Different countries, industries, and applications may use different units to represent the same physical quantity.

The **Unit Conversion Application** provides a simple way to convert a value from one unit to another. It is developed using Python and uses a menu-driven approach to make the application easy to operate.

The user can select a conversion category and provide the required

input. The program then applies the appropriate conversion formula and displays the result.

This project demonstrates how programming can be used to solve practical problems and perform calculations efficiently.

PAGE 7 – OBJECTIVES

OBJECTIVES OF THE PROJECT

The main objectives of the Unit Conversion Application are:

1. To develop a simple unit conversion application using Python.
 2. To provide multiple categories of unit conversion.
 3. To reduce the need for manual calculations.
 4. To provide quick and accurate conversion results.
 5. To understand the practical use of Python programming.
 6. To implement conditional statements and loops.
 7. To understand user input and output handling.
 8. To develop a simple menu-driven application.
 9. To improve problem-solving and programming skills.
 10. To create a useful beginner-level real-world project.
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PAGE 8 – TECHNOLOGIES AND TOOLS USED

TECHNOLOGIES USED

1. Python

Python is a high-level, interpreted programming language. It is easy to understand and widely used for developing different types of applications.

Python is used in this project to implement the conversion logic and user interface.

2. Visual Studio Code

Visual Studio Code is used as the development environment for writing, editing, and running the Python program.

3. GitHub

GitHub can be used to store and manage the project source code online. It also helps in maintaining and sharing the project repository.

4. Operating System

The application can be executed on a computer system with Python installed.

PAGE 9 – FEATURES OF THE APPLICATION

FEATURES

The Unit Conversion Application provides the following features:

1. Length Conversion

Converts units such as metre, kilometre, centimetre, millimetre, inch, foot, and mile.

2. Weight Conversion

Converts units such as kilogram, gram, milligram, pound, and ounce.

3. Temperature Conversion

Converts values between Celsius, Fahrenheit, and Kelvin.

4. Time Conversion

Converts seconds, minutes, hours, days, and other time units.

5. Area Conversion

Converts square metre, square kilometre, square centimetre, acre, hectare, etc.

6. Volume Conversion

Converts litre, millilitre, cubic metre and other volume units.

7. Speed Conversion

Converts kilometre per hour, metre per second, mile per hour, etc.

8. Digital Storage

Converts units such as byte, kilobyte, megabyte, gigabyte, and terabyte.

PAGE 10 — WORKING OF THE APPLICATION

WORKING / METHODOLOGY

The application follows a simple step-by-step process:

Step 1: Start the program.

Step 2: Display the main conversion menu.

Step 3: Ask the user to select a conversion category.

Step 4: Display the available units for the selected category.

Step 5: Ask the user to enter the value to be converted.

Step 6: Apply the appropriate conversion formula.

Step 7: Display the converted result.

Step 8: Allow the user to perform another conversion or exit the program.

Basic Flow

**START → SELECT CATEGORY → SELECT UNITS → ENTER VALUE →
CALCULATE → DISPLAY RESULT → CONTINUE/EXIT**

PAGE 11 – ALGORITHM

ALGORITHM

11. Start the program.
12. Display the title of the application.
13. Display the list of conversion categories.
14. Take the user's category choice as input.
15. Check the selected category.
16. Display the corresponding units.
17. Take the source unit and target unit.
18. Take the numerical value from the user.
19. Apply the required conversion formula.

20. Display the converted value.
 21. Ask whether the user wants to perform another conversion.
 22. If yes, return to the main menu.
 23. If no, terminate the program.
 24. End.
-

PAGE 12 — FLOWCHART

FLOWCHART OF THE PROJECT

Draw the following flowchart manually using boxes and arrows:

START



Display Main Menu



Select Conversion Category



Select Source & Target Units



Enter Value



Apply Conversion Formula



Display Converted Result

↓

Another Conversion?

↙ ↘

YESNO

↓ ↓

Main MenuEND

PAGE 13 — CONVERSION FORMULAS

IMPORTANT CONVERSION FORMULAS

Length

1 metre = 100 centimetres

1 kilometre = 1000 metres

1 foot = 0.3048 metre

1 inch = 0.0254 metre

Weight

1 kilogram = 1000 grams

1 gram = 1000 milligrams

1 pound ≈ 0.453592 kilogram

Temperature

Celsius to Fahrenheit:

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32$$

Fahrenheit to Celsius:

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$

Celsius to Kelvin:

$$\text{K} = ^{\circ}\text{C} + 273.15$$

Time

1 minute = 60 seconds

1 hour = 60 minutes

1 day = 24 hours

PAGE 14 — SAMPLE INPUT AND OUTPUT

SAMPLE OUTPUT

Example 1: Length Conversion

Input:

Enter value: 5

From: Kilometre

To: Metre

Output:

5 kilometre = 5000 metre

Example 2: Temperature Conversion

Input:

Enter temperature: 25 $^{\circ}$ C

Output:

25 $^{\circ}$ C = 77 $^{\circ}$ F

Example 3: Weight Conversion

Input:

Enter value: 2

From: Kilogram

To: Gram

Output:

2 kilogram = 2000 gram

You can also paste or draw **screenshots of your actual program output** on this page.

PAGE 15 – ADVANTAGES AND LIMITATIONS

ADVANTAGES

25. Easy to use.
26. Simple menu-driven interface.
27. Provides quick results.
28. Supports multiple conversion categories.
29. Reduces manual calculations.
30. Useful for students and beginners.
31. Demonstrates practical Python programming.

LIMITATIONS

32. It is mainly a console-based application.

- 33. The number of supported units is limited to the units implemented in the program.
 - 34. It does not provide a graphical user interface.
 - 35. It may require additional programming to support new units.
 - 36. It does not necessarily store conversion history.
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PAGE 16 — FUTURE SCOPE

FUTURE SCOPE

The Unit Conversion Application can be improved in several ways in the future.

- 37. A graphical user interface can be added.
 - 38. More units and conversion categories can be included.
 - 39. A mobile application can be developed.
 - 40. A web-based version can be created.
 - 41. Conversion history can be added.
 - 42. Error handling can be improved.
 - 43. More advanced scientific conversions can be included.
 - 44. The application can support multiple languages.
 - 45. A search option can be added to find units quickly.
 - 46. The project can be connected with an online database for additional conversion information.
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PAGE 17 – CONCLUSION

CONCLUSION

The **Unit Conversion Application** successfully demonstrates the practical application of Python programming.

The project provides a simple method for converting values between different units. It includes various conversion categories such as length, weight, temperature, time, area, volume, speed, and digital storage.

During the development of this project, important programming concepts such as variables, user input, conditional statements, loops, functions, and mathematical calculations were applied.

Overall, this project helped in developing programming, logical thinking, and problem-solving skills. It also demonstrates how Python can be used to create useful real-world applications.

PAGE 18 – REFERENCES

REFERENCES

47. Python Programming Language Documentation.
48. Visual Studio Code Documentation.
49. GitHub Documentation.
50. Basic Mathematics and Unit Conversion References.
51. Python programming notes and classroom material.

PROJECT DETAILS

Project Title: Unit Conversion Application

Programming Language: Python

Development Environment: Visual Studio Code

Repository: GitHub

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Academic Session: 2026– 27